Data Analytics Research assignment

1. Database

Is an organized, electronic collection of structured data stored on a computer system designed for efficient retrieval, management and updating. Common types include relational (SQL) for structured data and non-relational (NoSQL) for unstructured data

2. Data Lake

Is a centralized repository that stores vast amounts of raw, unstructured, semi-structured and structured data in its native format. Unlike data warehouse, it uses a "schema-on-read" approach allowing data to be ingested instantly without upfront modeling making it ideal for big data, AI and machine learning

3. Data warehouse

Is a centralized, specialized data management system designed to aggregate, store and analyze large volumes of historical data from multiple, disparate sources. Used for business analytics, complex reporting and gathering insights from multiple systems eg Snowflake, Amazon Redshift, Google BigQuery

4. Data Mart

Is a focused, subset of a data warehouse designed for specific business lines, departments or teams eg finance or marketing) It stores curated, non-volatile historical data, offering improved query performance, faster access to insights and enhanced security for specific users often serving as a "single of truth" for that department

5. Structured, Semi-structured and Unstructured data

- Structured data is highly organized and follows a rigid, predefined schema (eg excel spreadsheet
- Semi-structured data does not follow a strict tabular model but uses tags or markers to separate semantic elements and enforce hierarchies (eg email headers (To, From, Date)
- Unstructured data has no predefined format or organization, making it the most complex type to process. (eg Images

6 Data modeling

Is the process of creating visual representations or blueprints of an information system to define how data is collected, stored and used

Conceptual Data Model: Identifies the highest level relationships between different entities, focusing on business concepts rather than technical details

Logical Data Model: Adds more detail such as attributes, primary keys and relationships but remains independent of a specific database technology

7 Data mining

Is the process of using machine learning statistics and AI to analyze large datasets to uncover hidden patterns, trends, correlations and anomalies. Eg Retail and Mining

8 Database Management System (DBMS)

Is software that acts as an interface between users applications and data, allowing for the efficient creation, retrieval, updating and management of structured or unstructured data

* Relational DBMS
* Graph DBMS
* No SQL DBMS
* Network DBMS

9 ETL (Extract, Transform, Load)

Is a critical data integration process that retrieves raw data from multiple sources, cleans and converts it into a structured format and loads it into a centralized system like a data warehouse. It acts as a data pipeline, converting raw data into actionable information for analysis and business intelligence

10 CSV - comma separated values: Simple, human-readable text format, widely used for data exchange spreadsheets and databases

JSON: JavaScript Object Notation: Lightweight text-based format for data interchange, ideal for APIs and web services

Parquet: Apache column-oriented format optimized for big data offering high compression and fast query speeds

Avro: Row-based format specializing in schema evolution, data serialization and streaming

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11 Artificial Intelligence (AI)

Is a broad field of machines simulating human intelligence to analyze, predict or classify data such as recommendation engines.

Narrow AI is designed to excel at specific, predefined tasks such as facial ~~reg~~ recognition or language translation and represents all AI technology

General AI is a theoretical, human-like intelligence capable of learning, reasoning and applying knowledge ~~over~~ across any domain which does not yet exist.

12 Machine Learning (ML)

Is a subset of artificial intelligence (AI) focused on building system that learn patterns from data to make predictions or decisions without being explicitly programmed for every rule

Supervised Learning: Algorithms learn from labeled data paired inputs and desired outputs to classify data or predict outcomes

Unsupervised Learning: Algorithms analyze unlabeled data to find hidden structures patterns or groupings (clustering).

Reinforcement learning: Agents learn to make decisions by taking actions in an environment to maximize a reward often used in robotics and AI games

13 Deep Learning

Is a subset of machine learning based on artificial neural networks with multiple layers (hence "deep") that mimic the human brain to learn patterns from vast amounts of data

14 Natural Language Processing (NLP)

Is technology that allows to interpret, manipulate and comprehend human language

11 different types of techniques: Statistical Techniques, Stochastic Techniques, Rule Based Techniques and Hybrid Techniques.

15 Large language model (LLMs)

Are a type of artificial intelligence designed to understand and generate human-like text based on the input they receive. ~~An~~ An LLM is a computer program that has ~~read~~ enough ~~examples~~ ~~examples~~ to be able to recognize and interpret human language or other types of complex data

Many LLMs are trained on data that has been gathered from the Internet, thousands or millions

of gigabytes worth of text

16 Generative AI

Is a type of artificial intelligence that creates new, original content such as text, images, music or code by learning patterns from existing data.

Common Use Cases

Text generation - Drafting emails, writing essays or creating marketing content.

Image generation - Creating art or photo realistic images from text descriptions.

Coding - Writing, debugging and translating code.

17 Training data

Is the initial dataset used to teach AI and machine learning models to recognise patterns, make predictions or generate content. It acts as a foundational reference (often 70-80% of total data) that allows algorithms to adjust internal parameters to minimize errors. High-quality training data ensures reliable AI performance.

18 Computer Vision

Is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and visual inputs, allowing them to take action or make recommendations. Eg Image Recognition and Object Detection.

19 API (Application Programming Interface)

Is a set of rules, protocols and definitions that allows different software applications to communicate and exchange data with each other. It acts as an intermediary enabling one piece of software to use the functions or data of another without needing to know how that underlying code is built.

20 API keys

Are unique alphanumeric strings used to authenticate and authorize applications or users when calling an API. They act as digital security tokens identifying the requester to manage access, track usage and enforce rate limits. API keys are crucial for enabling server to server communication or embedding services like maps or AI in apps.

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21 Rest API

Is an ~~adjacent~~ architectural style for designing networked applications. It allows different software systems such as a mobile app and a web server to communicate over the internet using the Standard HTTP Protocol

22 JSON API

Is a web service that uses JavaScript Object Notation (JSON) as its primary data exchange format. While the term often refers to any API using JSON it frequently specifically refers to the JSON API specification, a set of conventions for building consistent and efficient APIs

23 Prompt engineering

Is the art of designing, refining and optimizing inputs (prompts) to guide AI models like LLMs to produce accurate, relevant and high-quality outputs. It involves providing specific context instructions and examples to improve performance often acting as a "programming in natural language" to unlock model capabilities and reduce errors.

24 Map function

Is a fundamental tool in programming used to transform a collection of items (like a list or array) by applying a specific function to every individual element within that collection

JavaScript - It is frequently used for data transformation and rendering lists in frameworks like React

Python - The built-in map() function returns an iterator (in Python 3) or a list (in Python 2)

25 Cloud computing

Is the on-demand delivery of IT resources such as storage servers and software over the internet with pay as you go pricing, allowing users to avoid owning physical data centers

Types of Cloud Services

IaaS Provides foundational IT infrastructure like virtual servers and storage

PaaS - Delivers a framework for developers to build test and deploy applications

SaaS - Delivers software applications over the internet on a subscription basis

26. Version Control System (VCS)

It's a systems tool used to capture development to track, manage and record changes to source code over time.

A commit is a recorded snapshot of changes made to the codebase at a specific point in time.

A branch is a separate line of development that diverges from the main code.

27. Jupyter Notebook

It's an open-source web application that allows users to create and share documents containing live code, equations, results, rich text and multimedia.

Data science: Exploratory data analysis (EDA), data cleaning, statistical modelling.

28. Python

It is a general purpose language that supports multiple paradigms, including object oriented, procedural and functional programming.

Numpy - Known as "Numerical Python" it is the foundational library for scientific computing.

Pandas - A critical tool for data analysis and manipulation.

Matplotlib - The standard library for data visualization.

Scikit learn - A premier library for machine learning.

29. Data privacy

Is the right of individuals to control how their personal information is collected, stored and used by organizations. It focuses on managing data transparency, consent and security to prevent unauthorized access or misuse, regulated by laws like GDPR and South Africa POPIA.

30. Data ethics

Refers to the moral obligations and principles guiding the responsible collection, protection, analysis and usage of data particularly personally identifiable information (PII).

31. Data governance

Is a discipline that manages the availability, usability, integrity and security of an organization's data through defined policies, standards and roles.

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Core Pillars

* Roles and Responsibilities
* Data lifecycle management

32. AI bias

Refers to systematic, unfair errors in AI system output caused by prejudiced training data, flawed algorithm design or human prejudices

Real World Consequences

Face Recognition - Often struggles with misidentifying darker skinned individuals

Hiring - Can favor male applicants over females due to historical data

Language Models - May associate specific professions with gender stereotypes

33. Data Engineer (The Architect)

Focuses on creating data pipelines, managing data warehouses and ensuring data accessibility. They use SQL, ETL tools and Big Data technologies to build the infrastructure that other roles rely on

Data Analyst (The Reporter)

Focused on interpreting historical data to find trends, create visualizations and generate business insights. They primarily use SQL, Python and BI tools (eg Tableau, Power BI) to report on 'what happened'

Data Scientist (The Researcher)

Focuses on building predictive models, experiments and advanced algorithms (machine learning) to forecast future outcomes. They use Python, statistics and machine learning libraries (eg scikit-learn)

AI Engineer (The Builder)

Focuses on applying and deploying AI models, LLMs and agents into production systems that end-users interact with. They work with Deep learning frameworks (Tensorflow, PyTorch) and API integrations

34. Business Intelligence (BI)

Encompasses technologies, strategies and processes used to analyze organizational data, transforming raw information into actionable insights for better decision making. The BI process involves collecting, managing and analyzing data through tools such as Microsoft Power BI, Google Cloud, Looker and other reporting applications

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35 Generative AI. I'm learning more and more about AI. Becoming more interesting

36 API but I used google to understand it a bit bit more

37 Yes. Goes hand in hand with keeping a Data analysis

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